

A Finite Support-Closure Orbit Atlas for a Named Abelian Core

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Abstract

We enumerate the complete support geometry of sixteen frozen named coordinates in the finite abelian core $Q \cong \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2$. A support is a subset of the labels and its closure is the subgroup generated by the selected coordinates. The 11520-element C75 lifted symmetry has an order-six kernel on labels, so the faithful effective action is generated by five explicit permutations and has order 1920. Its action partitions all $2^{16} = 65536$ supports into 3024 orbits. Exact closure tracking finds 98 single-deletion-minimal supports in 34 orbits; imposing closure equal to all of Q leaves 25 supports in seven orbits, all of cardinality three. The computation is source-bound to the C75 evidence and records the ambient kernel explicitly. It is a finite named support result, with no arithmetic, local, Euler-factor, root-number, Burnside-ring, or Hilbert–Polya claim.

1 Finite support object

Let

$$Q = \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2$$

and let $(x_i)_{i=1}^{16}$ be the frozen C75 named coordinates. Write $L = \{S_1, \dots, S_{16}\}$ for the labels. For a support $A \subseteq L$, define

$$\Phi(A) = \langle x_i : S_i \in A \rangle \leq Q, \quad \Phi(\emptyset) = \{0\}.$$

The complete support space therefore has $2^{16} = 65536$ elements. C76 records both the generated subgroup and the orbit of each support; it does not sample supports or replace the named presentation by an abstract isomorphism type. The scope firewall is NO_BAD_EULER_OR_ROOT_NUMBER.

2 The effective label group

The C75 lifted object retains an ambient automorphism in each pair and has order 11520. Its ambient order-six factor is a kernel for the induced action on the sixteen labels. C76 removes only this action kernel and works with the faithful permutation group $H \leq \text{Sym}(L)$ generated by

$$\begin{aligned} z_5 &= (S_5 \ S_6 \ S_{10} \ S_{13} \ S_{14}), \\ z_2 &= (S_5 \ S_6), \\ a &= (S_3 \ S_{11}), \quad c = (S_7 \ S_{15}), \\ u &= (S_1 \ S_{16})(S_3 \ S_4)(S_{11} \ S_{12}). \end{aligned}$$

Theorem 1 (Effective label action). *Direct permutation closure gives $|H| = 1920$. Its element-order distribution is*

order	1	2	3	4	5	6	10	12	20
count	1	311	20	584	24	460	264	160	96.

GAP identifies the compatible direct-product structure as $C_2 \times S_5 \times D_8$ (equivalently $S_5 \times C_2 \times D_8$).

The C6 factor is therefore a recorded boundary condition, not an additional generator of support orbits. This distinction is essential: using the 11520-element ambient lift would collapse to the same support partition but would misstate the effective group and its stabilizers.

3 Complete support orbit atlas

The effective group acts on supports by

$$h \cdot \{S_i : i \in A\} = \{h(S_i) : S_i \in A\}.$$

The producer and checker enumerate every orbit and choose its smallest mask as representative.

Theorem 2 (Orbit partition). *The 65536 supports form exactly 3024 H -orbits. Their size spectrum is*

$ O $	1	2	4	5	8	10	16	20	40	80	160
$\#O$	128	256	416	128	192	384	16	672	608	208	16.

The number of orbits by support cardinality $k = 0, \dots, 16$ is

$$1, 7, 27, 73, 151, 252, 352, 424, 450, 424, 352, 252, 151, 73, 27, 7, 1.$$

The size-weighted orbit sum is 65536.

The closure atlas contains all twenty subgroups of Q . Aggregating rows by subgroup order gives the following independent view of the same partition:

$ \Phi(A) $	1	2	3	6	9	18	27	54
#supports	32	32	256	256	2080	2080	30400	30400
#orbits	6	6	24	24	150	150	1332	1332.

The evidence keeps the finer subgroup-index counts, so this aggregation does not discard distinctions between subgroups of equal order.

4 Minimal supports

Call A closure-minimal when $A = \emptyset$ or

$$\Phi(A \setminus \{S_i\}) \neq \Phi(A) \quad \text{for every } S_i \in A.$$

This is a single-deletion definition and is checked directly for all 65536 supports.

Proposition 1 (Closure-minimal atlas). *There are 98 closure-minimal supports, with cardinality counts*

$$\{0 : 1, 1 : 11, 2 : 48, 3 : 38\}.$$

They form 34 H -orbits with spectrum

$$\{1 : 10, 2 : 8, 4 : 14, 8 : 2\}.$$

Among them, 25 supports generate the full 54-point core Q . Every such support is a triple; the 25 supports form seven orbits with spectrum

$$\{1 : 1, 2 : 2, 4 : 3, 8 : 1\}.$$

Their canonical representative masks are

$$261, \quad 265, \quad 321, \quad 324, \quad 385, \quad 448, \quad 33025.$$

The masks are retained rather than replaced by informal label descriptions so that the result can be compared byte-for-byte with the canonical evidence. For example, with the least significant bit assigned to S_1 , mask 261 is the support $\{S_1, S_3, S_9\}$ and mask 33025 is $\{S_1, S_9, S_{16}\}$.

5 Reproducibility and scope

The source-bound producer and independent checker recompute the subgroup extension table, effective group, orbit partition, closure atlas, and both minimality filters. GAP, clean replay, and hostile semantic mutation tests are separate gates; they must preserve the C6-kernel boundary and the exact source hashes. The resulting object is an exact finite support-orbit atlas for the named coordinates. It is not a full table of marks or Burnside-ring calculation, and it carries no arithmetic/local, Euler-factor, root-number, automorphy, or Hilbert–Polya interpretation.